Individual Contribution Report

SurfWatch — AI-Powered Rip Current Detection System

Student Name:	**Alan Bello**	Student ID:	**AlanMBello**
Subject:	**COMP6002**	Sprint:	**SCRUM Sprint 1**
	SCRUM-20 — Create Dataset Card		

1 INTRODUCTION

SurfWatch is a university group project developed under the COMP6002 subject, aimed at building an AI-powered rip current detection system for beach safety. The system is designed to monitor shore-based camera feeds in real time and alert lifeguards and beachgoers to the presence of dangerous rip currents. The project follows an Agile Scrum methodology, with work organised into sprints tracked via Jira and a shared GitHub repository.

My role within the SurfWatch team is as a contributor to the Dataset and Data Preparation workstream (SCRUM-18). During Sprint 1, I was assigned SCRUM-20 — the task of researching and documenting a dataset card for the rip-current dataset(s) selected for training the SurfWatch MVP model. This report reflects on my individual contributions, the Agile tools I used, technical decisions made, and lessons learned throughout the process.

2 OVERVIEW OF CONTRIBUTIONS

My primary contribution during Sprint 1 was completing SCRUM-20: Create Dataset Card. This task sat under the parent epic SCRUM-18 (Dataset and Data Preparation) and required me to research, evaluate, and document a suitable rip-current dataset for use in training the SurfWatch MVP detection model. The deliverable was a structured dataset card committed to the project repository at **docs/datasets.md**.

Key contributions included:

- **Dataset Research:** Identified and evaluated the RipVIS v1.8.4 dataset on HuggingFace (https://huggingface.co/datasets/Irikos/RipVIS) as the primary dataset for SurfWatch MVP, based on its domain relevance, annotation quality, and format compatibility.
- **Dataset Card Creation:** Authored a comprehensive docs/datasets.md covering source, licence, size, annotation type, viewpoint diversity, strengths, limitations, and training suitability.
- **PDF Documentation:** Produced a styled PDF version of the dataset card and a fillable PDF form template for future dataset onboarding by the team.
- **Version Control:** Set up Git on my local machine, cloned the surf-watch repository, created a feature branch from develop, committed the dataset card, and opened Pull Request #10 for peer review.

These contributions directly support the project goal of building a robust, well-documented training pipeline for the SurfWatch rip current detection model. Selecting and documenting the right dataset is a foundational step that enables all downstream model development work.

3 AGILE TOOLS UTILISED

- **Jira (Task Board & Sprint Tracking):** SCRUM-20 was managed entirely through the SurfWatch Jira board. The task was assigned to me with a Medium priority, linked to parent epic SCRUM-18, and tracked through the sprint. I used the Jira ticket to understand the acceptance criteria — specifically that the dataset card must be created in the repo, include source link, licence, size, annotation type, diversity notes, and known issues, and clearly state why the dataset fits SurfWatch MVP. Once the PR was submitted, I updated the ticket status to In Review and left a comment with the PR link.

• **GitHub (Version Control & Pull Requests):** The team's source code and documentation is hosted on GitHub at https://github.com/COMP6002-SurfWatch/surf-watch. I used Git to clone the repository, create a feature branch (feature/SCRUM-20-dataset-card) from the develop branch, commit the dataset card, and push the branch. Pull Request #10 was opened targeting the develop branch, with Yuvindu Rashmike Caldera requested as reviewer.

• **WhatsApp (Team Communication):** The team used WhatsApp for informal coordination. Key decisions communicated through the group chat included clarification that the dataset card should be submitted as a report (not a new dataset card from scratch), and that the branch must be created from develop, not main. This was critical guidance that shaped my approach to the task.

• **Sprint Retrospective:** While a formal sprint retrospective had not yet been completed at the time of writing, the sprint board showed clear progress against the backlog, with SCRUM-20 moving from To Do → In Progress → In Review following my PR submission.

4 TECHNICAL CONTRIBUTIONS

My technical work for SCRUM-20 spanned three areas:

4.1 Dataset Research and Evaluation

I researched publicly available rip current datasets and identified RipVIS v1.8.4 as the strongest candidate for SurfWatch MVP. RipVIS was introduced alongside the paper "RipVIS: Rip Currents Video Instance Segmentation Benchmark for Beach Monitoring and Safety" (CVPR 2025) and is hosted on HuggingFace. Key evaluation criteria included:

• Domain alignment — shore-facing video of rip currents, directly matching SurfWatch's deployment context

• Annotation format — COCO JSON and YOLO .txt formats provided, compatible with YOLOv8/YOLO11 pipelines

• Expert-validated split — manual train/val/test split by rip current domain specialists

• 184 videos (150 rip current, 34 no-rip), plus 2,466 additional training images from a 2023 CVPRW predecessor paper

• Licence: CC BY-NC 4.0 — suitable for academic MVP development with written permission required for commercial deployment

4.2 Dataset Card Authoring (docs/datasets.md)

I authored a structured Markdown dataset card covering all fields required by the acceptance criteria. The card was written to serve as a living reference document for the team, formatted consistently with other docs/ files in the repository. Sections included: Source, Licence, Size, Annotation Type, Viewpoint Diversity, Strengths, Limitations and Known Issues, Training Suitability for SurfWatch MVP, and a BibTeX citation block.

4.3 PDF Documentation

I produced two supporting PDF documents: a styled read-only PDF rendering of the dataset card, and a fillable PDF form designed for the team to use when onboarding future datasets. The fillable form included interactive text fields, checkboxes, and radio buttons covering all dataset card sections, enabling consistent documentation across datasets as the project scales.

4.4 Git Workflow

I set up Git on my local machine (installing Git 2.53.0 for Windows), cloned the surf-watch repository, checked out the develop branch, and created the feature branch feature/SCRUM-20-dataset-card. After committing the dataset card with the message "SCRUM-20: Add RipVIS dataset card", I pushed the branch to origin and opened Pull Request #10 on GitHub targeting develop.

5 COLLABORATION AND COMMUNICATION

- **Jira Coordination:** I communicated task progress through the Jira ticket by updating the status and posting a detailed comment on SCRUM-20 once the PR was open, including the PR link and a summary of what was changed and how it was tested.
- **WhatsApp Team Chat:** I used the team WhatsApp group to clarify the task scope with Yuvindu Rashmike Caldera. This was particularly important for two decisions: (1) whether to create a new dataset or just the report — the team confirmed just the report was needed; and (2) confirming the correct base branch for my feature branch was develop, not main.
- **GitHub Pull Request:** I opened PR #10 and requested Yuvindu as reviewer, using the team's standard PR template which includes sections for Jira ticket reference, what changed, how it was tested, screenshots, and a checklist of acceptance criteria.
- **Reviewer Assignment:** I assigned myself as the PR assignee and requested review from the relevant team member, keeping the workflow transparent and traceable for the whole team.

6 PROBLEM-SOLVING AND DECISION MAKING

- **Dataset Selection:** The task did not specify which dataset to use, requiring me to independently research and evaluate options. I decided on RipVIS v1.8.4 based on its CVPR 2025 peer-reviewed status, domain specificity, and YOLO format compatibility with the team's likely training pipeline. I documented this decision transparently in the dataset card's Training Suitability section, including both strengths and limitations.
- **Licence Risk Identification:** During research I identified that RipVIS is licensed under CC BY-NC 4.0, which restricts commercial use. I proactively flagged this as a [redacted] risk in the dataset card and recommended the team seek written permission from the authors before any commercial deployment of the trained model.

• **Branch Target Correction:** I initially began to open the PR targeting main, but upon reviewing the team chat instructions and cross-checking the Jira ticket, I corrected this to target develop before submitting, ensuring compliance with the team's branching strategy.

• **Git Setup:** Having not used Git on my current machine, I encountered a "git not recognised" error. I resolved this by downloading and installing Git 2.53.0 for Windows, then configuring my global user identity before making the first commit.

7 CHALLENGES FACED

• **Unfamiliarity with Git:** Setting up and using Git via the command line for the first time on this machine presented a learning curve. Steps such as configuring user identity, understanding branch management, and pushing to a remote repository required careful guidance. This challenge was overcome by working through each step methodically, and the experience significantly improved my Git confidence.

• **Ambiguous Task Scope:** The initial Jira ticket did not explicitly specify which dataset to document, requiring me to make an independent judgment call on dataset selection. I addressed this by directly researching publicly available rip current datasets and confirming the scope with my teammate via the team chat.

• **Licence Complexity:** The CC BY-NC 4.0 licence on RipVIS includes additional restrictions beyond the standard Creative Commons terms, particularly around redistribution and commercial use. Carefully reading and interpreting these terms to accurately document them in the dataset card was a non-trivial research task.

• **PR Description Template:** The team's PR template was partially pre-filled but required manual completion. Some fields such as "How was it tested?" were not straightforward for a documentation task, requiring me to adapt the format appropriately.

8 LESSONS LEARNED

• **Git Workflow in a Team Context:** This task gave me practical experience with the full Git workflow — cloning, branching, committing, pushing, and opening pull requests. I now understand the importance of branching from the correct base branch and how PRs enable code review and quality control in a team environment.

• **Importance of Licence Review:** Dataset licences can have significant implications for how a model trained on that data can be deployed. This task reinforced the importance of checking licence terms early in the data selection process, particularly in projects that may transition from academic to commercial use.

• **Documentation as a First-Class Deliverable:** Producing the dataset card highlighted that documentation is as important as code in a machine learning project. A well-structured dataset card helps the whole team understand what data is available, its constraints, and how to use it correctly.

• **Clarifying Requirements Early:** The WhatsApp exchange about task scope demonstrated the value of clarifying ambiguous requirements early. Confirming that only a report was needed (not a new dataset card from scratch) saved significant time.

9 SELF-ASSESSMENT

Overall, I am satisfied with my contribution to SCRUM-20. I delivered all three acceptance criteria — the dataset card file in the repository, inclusion of all required fields, and a clear statement of training suitability — and submitted the PR within the sprint. I proactively identified a licence risk and communicated it clearly in the documentation.

Strengths	Thorough dataset research and evaluation; proactive risk identification (licence); complete and well-structured documentation; successful independent Git workflow setup
Areas for Improvement	Could have clarified task scope and branching strategy at the start of the sprint rather than mid-task; PR description could have been more detailed in the testing section
Agile Practices	Followed Agile practices well — used Jira for task tracking, communicated progress via team channels, and submitted work for peer review via pull request
Technical Growth	Significantly improved Git proficiency and gained practical experience with GitHub PR workflows and branch management in a team environment

10 TEAM FEEDBACK

As the PR (#10) was opened at the time of writing this report and is currently awaiting review from Yuvindu Rashmike Caldera, formal peer review feedback has not yet been received. However, the following informal feedback was communicated through the team WhatsApp group:

- Confirmation that the RipVIS HuggingFace dataset was the correct dataset to document for this task.
- Guidance on the correct branching strategy (develop, not main), which I incorporated into my workflow.
- Confirmation that the task required a report/dataset card document rather than a new dataset.

I anticipate that the formal PR review will provide additional feedback on the content and structure of the dataset card, which I will incorporate before the PR is merged into develop.

11 FUTURE GOALS

- **Expand Dataset Coverage:** In future sprints, I aim to contribute to evaluating and documenting additional supplementary datasets that could augment the RipVIS training data, particularly datasets with different beach environments and lighting conditions to improve model generalisation.
- **Deepen Git and GitHub Proficiency:** Having set up Git for the first time during this sprint, I intend to continue practising branching, rebasing, resolving merge conflicts, and contributing to code reviews to become a more effective collaborator on the team.
- **Contribute to Model Training Pipeline:** Now that the dataset has been selected and documented, I would like to contribute to subsequent SCRUM-18 subtasks related to data preprocessing and the training

pipeline setup, applying the dataset knowledge I have built during SCRUM-20.

• **Improve Agile Estimation:** I underestimated the time required for Git setup during this sprint. In future sprints, I will factor in environment setup time when estimating task effort to provide more accurate sprint forecasts.

12 CONCLUSION

During Sprint 1 of the SurfWatch project, I successfully completed SCRUM-20 — the creation of a dataset card for the RipVIS v1.8.4 rip current dataset. The deliverable, committed to docs/datasets.md on a feature branch from develop, satisfies all three acceptance criteria defined in the Jira ticket and provides the SurfWatch team with a comprehensive reference document for the primary training dataset.

This task gave me valuable experience across multiple dimensions: dataset research and evaluation, licence compliance analysis, technical documentation, and hands-on use of Git and GitHub in a team Agile workflow. The use of Jira for task tracking, GitHub for version control and peer review, and WhatsApp for team communication demonstrated how Agile tools work together to keep a distributed team aligned and productive.

The key lesson from this sprint is that foundational data documentation work is critical to the success of any machine learning project. By clearly documenting the dataset's source, licence, size, annotations, limitations, and suitability, SCRUM-20 ensures the entire team can make informed decisions about data usage throughout the project lifecycle. I look forward to building on this contribution in future sprints as SurfWatch progresses toward a fully functional MVP.

■ ***SCRUM-20 Acceptance Criteria Met: Dataset card created at docs/datasets.md · Includes source link, licence, size, annotation type, diversity notes, known issues · Clearly states why RipVIS fits SurfWatch MVP · PR #10 open for review on develop branch***